

Electrons

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1:02 PM

Energy	Dmax	Rp	R80	R90	Surface Dose	Rules of Thumb
6MeV	1.5 cm	3.5 cm	2.5 cm	1.8 cm	87%	3%/cm
10MeV	2.5 cm	5.0 cm	3.5 cm	2.7 cm	85%	2.5%/cm
15MeV	3.5 cm	7.0 cm	5.0 cm	4.0 cm	82%	2%/cm
18MeV	4.0 cm	8.5 cm	6.0 cm	4.5 cm	80%	2%/cm

Electron scattering:

- Elastic (no loss in energy)
- Inelastic (loss of energy)
- Bremsstrahlung (Electron swoops around nucleus, ejecting photon).
- Ionization
- Excitation.
- Atomic collisions (Fission, electron breaks up nucleus).

Rules of Thumb:

- $R_p = \text{MeV}/2$
- $R_{50} = \text{MeV}/2.33$
- $R_{80} = \text{MeV}/2.8$
- $R_{90} = \text{MeV}/3.2$
- $D_{\text{max}} = 1/2 E^{2/3}$
- Beam divergence = $(f + z)^2 / (f)$
- Energy at depth: $E_z = E_0 (1 - z/R_p)$ where z is the depth and E_0 is the average energy.

Basic Concepts:

- **Surface dose** increases with energy, obliquity, and smaller fields.
- **Dmax** increases with **energy** and **field size** (taking longer to reach LSE).
- **Lower energy beams are flatter**, while higher energy beams are more forward scattered and less flat (curving inwards).
- Increased SSD creates greater beam divergence, beam gets less flat.
- Effective SSD with airgap = $l_o/l_g = ((\text{SSD} + d_{\text{max}} + \text{gap})/(\text{SSD} + d_{\text{max}}))^2$
- Absolute dose increases with field size due to increased collimator scatter (less relevant for high energies).
- Most probably energy decreases linearly with depth, $E_z = E_0(1 - z/R_p)$.

Beam irregularities:

- When going into lung, dose drops first (loss of scatter), but has a longer tail than if the beam were going through water.

- Hot spots are found inside a cavity and outside a bolus.
- High density materials introduce additional scattering beyond the small inhomogeneity.
- Photon/electron fields have a hot spot in the photon, and cold spot in the electron field.

Bolus:

- Used to flatten irregular fields.
- Need to be careful about bolus edges.
- Decelerator (low energy material like Lucite) will reduce the average electron energy.

Blocks:

- Thickness of lead = $\text{MeV}/2 + 1 \text{ mm}$ (for safety)
- Thickness of Cerrobend = $1.2 \times \text{Thickness of lead}$.
- Blocks can reduce dose rate for small field sizes.

Internal Shield:

- Will reduce dose to area beyond shield, but will heat up tissue directly in front of it.
- Electron Back Scatter = $1 + 0.735e^{(-0.052 Ez)}$
- Backscatter for 5 MeV is 56%.
- 10 mm to 10% transmission of back scatter.
- Shielding thickness = $\text{MeV}/2 + 1\text{mm}$ for lead in mm and $1.2 \times$ for Cerrobend.

Beam characteristics:

- **Fluence factor** = $1/\cos(\theta)$
- **Uniformity index** = $\text{Area of 90\% isodose at reference} / \text{Area of 50\% isodose}$.
 - Reference depth is recommended to be $R_p/2$.
 - Uniformity > 0.7
- **Beam Flatness** = At 95% isodose line, variation should not exceed 5% (3% ideally) within 2cm from geometrics edge.
- **Beam Symmetry** = Points should be $< 2\%$ to symmetric.

Methods of modeling:

- **Square root method:**
 - Approximate central axis depth dose: $D_{xy} = \sqrt{D_{xx} * D_{yy}}$

Common Example Problems:

- 1) Energy left in a beam after a given depth.
 - a. For 9MeV beam, how much is left after 2 cm?
 - b. $E_z = E_0 (1 - z/R_p)$
 - $R_p = \text{MeV}/2, 4.5$
 - Energy used = 2 cm/4.5 cm total
 - Energy remaining = $1 - 2/4.5 = 37\%$
 - According to Figure 14.40, you can drop to 10% intensity in 10 mm of water for a 6 MeV water.